Sustainability

Standardizing Data Center Operational Carbon Tracking: Supplier Metrics vs. Operator Needs

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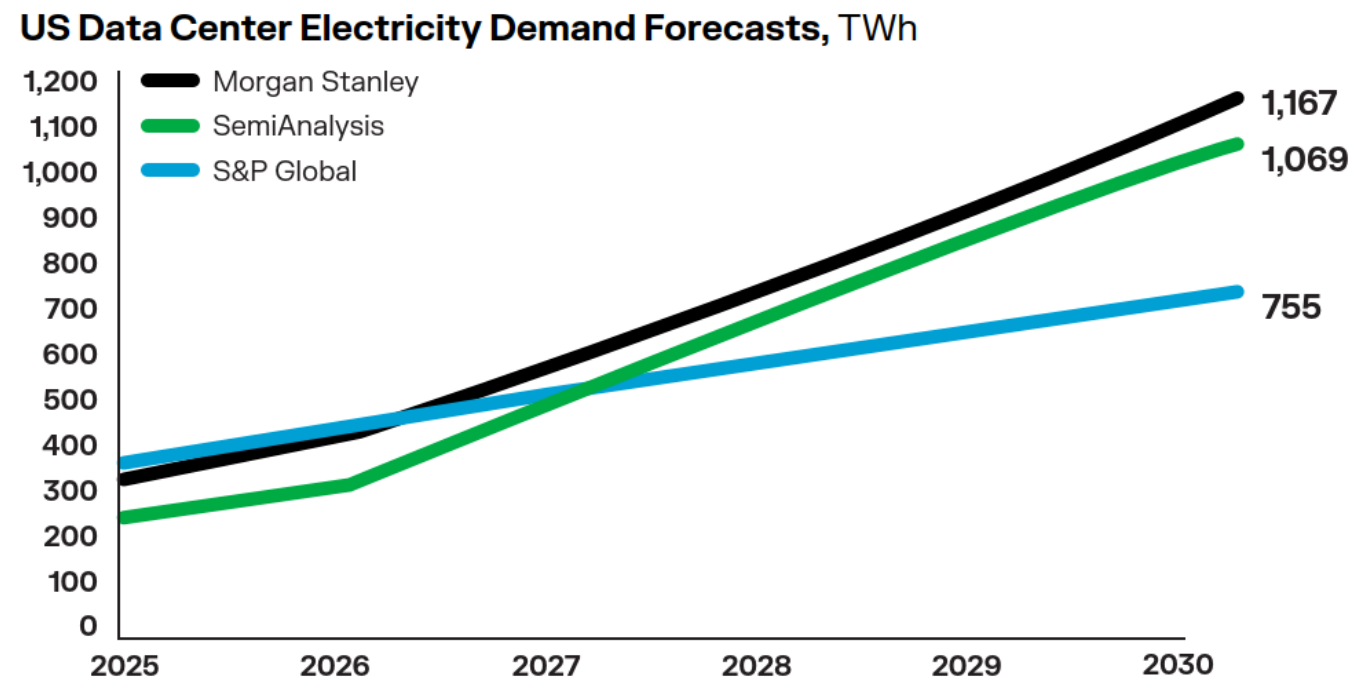
OCP SILVER Member

Agenda

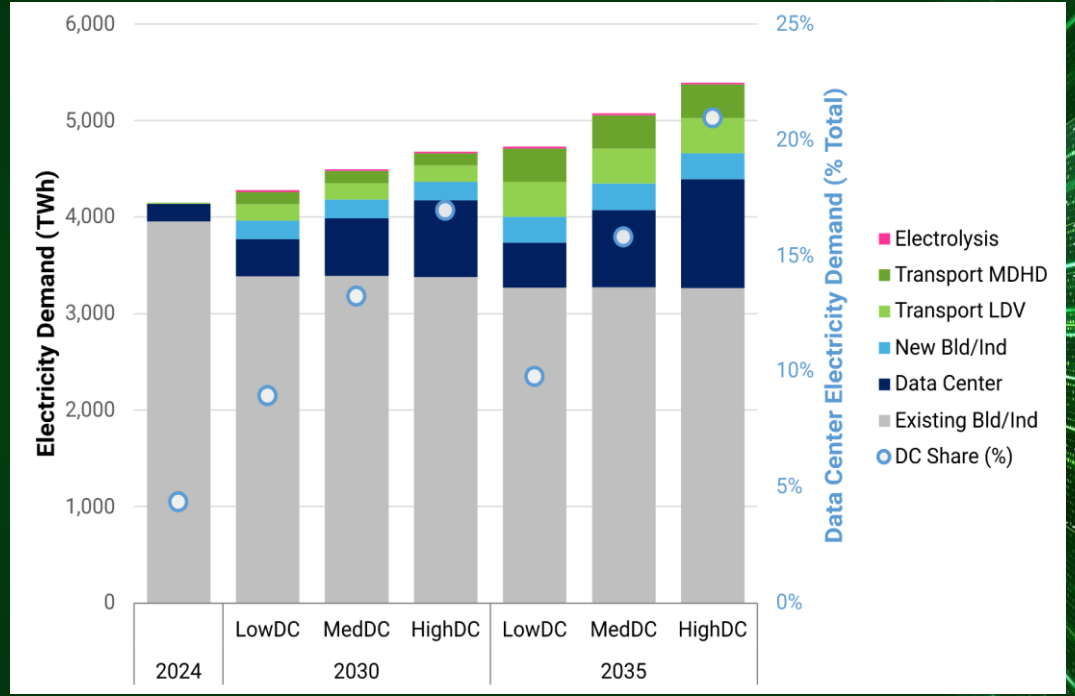
- Global data center growth outlook
- Data center carbon footprint lifecycle impact
- Bridging the information gap between DC operators and suppliers
- Harmonization frameworks and standards



Data Center Growth Forecast

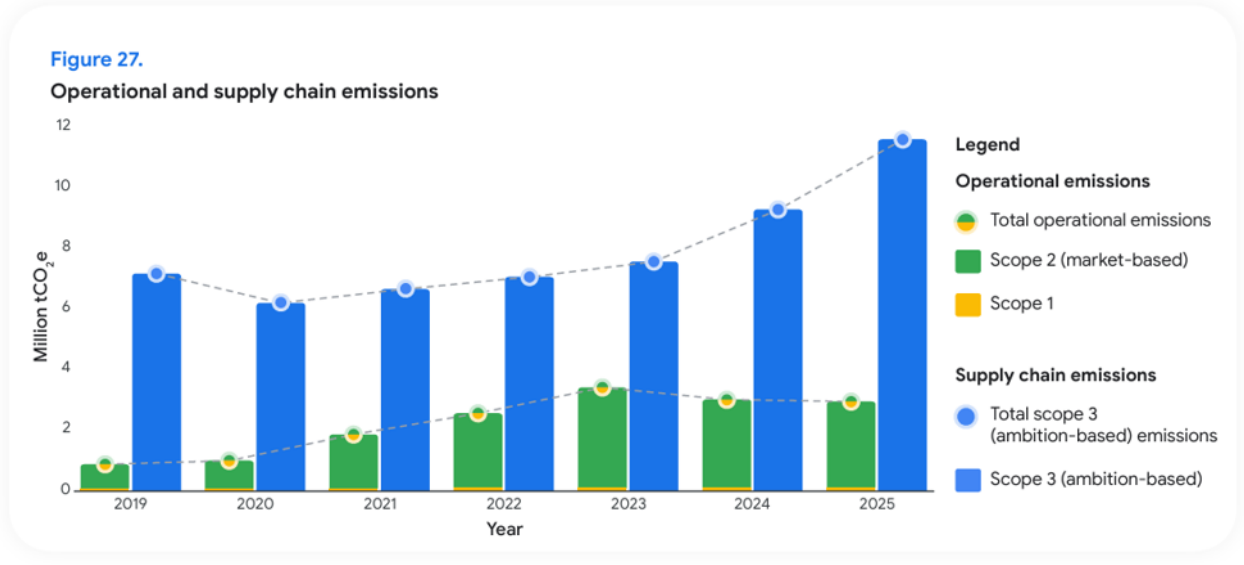


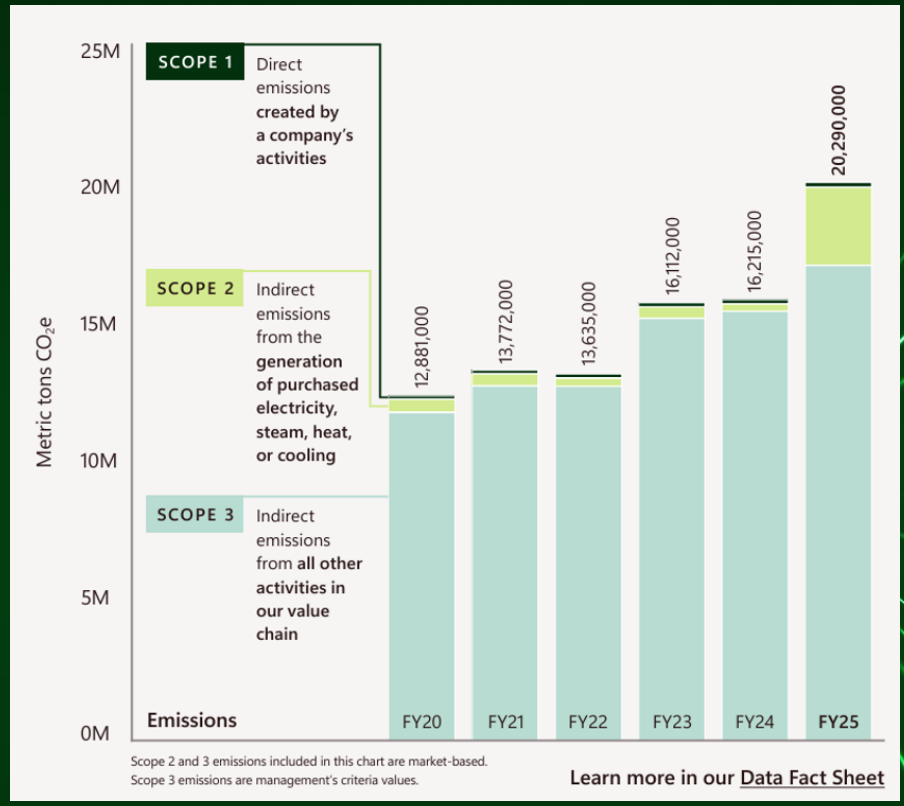
Bloom Energy (2026), 2026 Data center power report: the new realities shaping AI buidout



Powering Intelligence (2026), Data center load growth in context: https://powering-intelligence.epri.com/load-growth.html

Hyperscalers' Sustainability Commitment





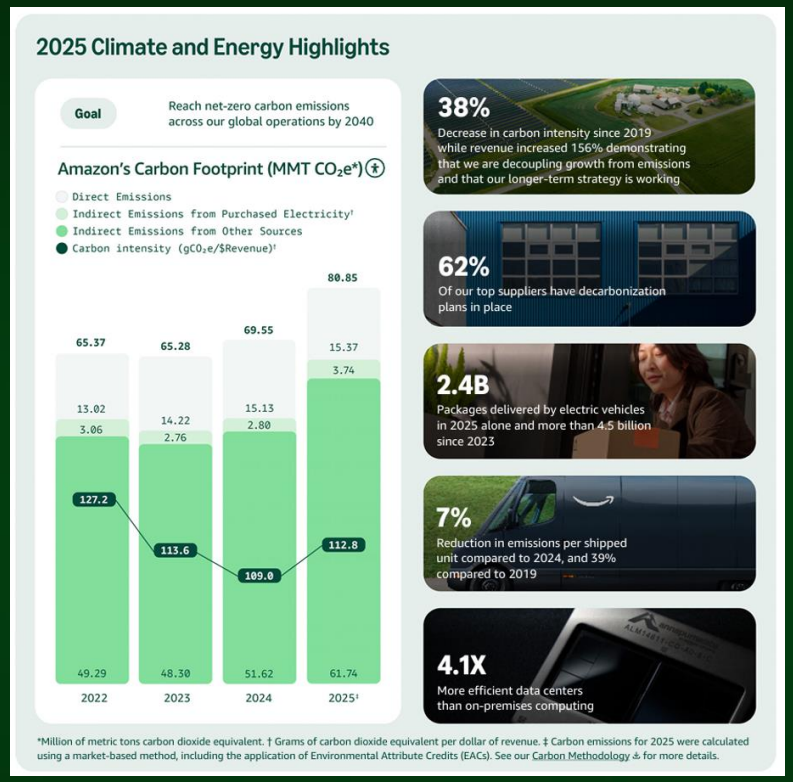
- Google Sustainability Goal: achieve net-zero across operation and value chain by 2030 and run on 24/7 carbon-free energy everywhere.

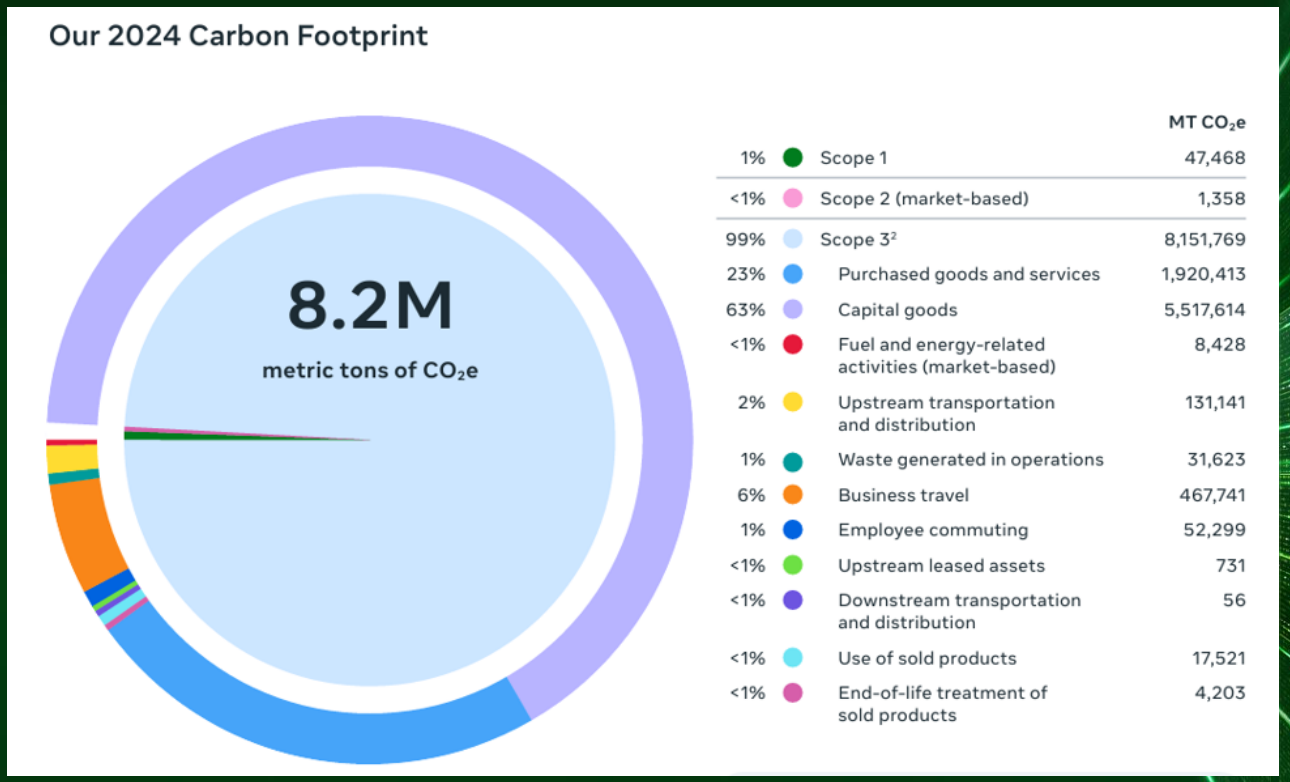
Google 2026 Environmental Report

- Microsoft Sustainability Goal: become carbon negative, water positive, and zero waste by 2030.

Microsoft 2026 Environmental Report

Hyperscalers' Sustainability Commitment





- Amazon Sustainability Goal: pledge to reach net-zero carbon across its business by 2040.

Amazon 2025 Environmental Report

- Meta Sustainability Goal: become carbon negative, water positive, and zero waste by 2030.

Meta 2025 Environmental Report

Carbon Accounting Methodologies

Operator Mandates

Data center operators face stringent regulatory pressure (EU CSRD, CSDDD, GHG Protocol) requiring granular accounting of Scope 1, 2 and Scope 3 operational greenhouse gas emissions across facility lifecycles.

Supplier Reliance

To quantify equipment impacts, operators rely heavily on Environmental Product Declarations (EPDs) and Life Cycle Assessments (LCAs) from major vendors of ICT, capital goods, EMP, etc..

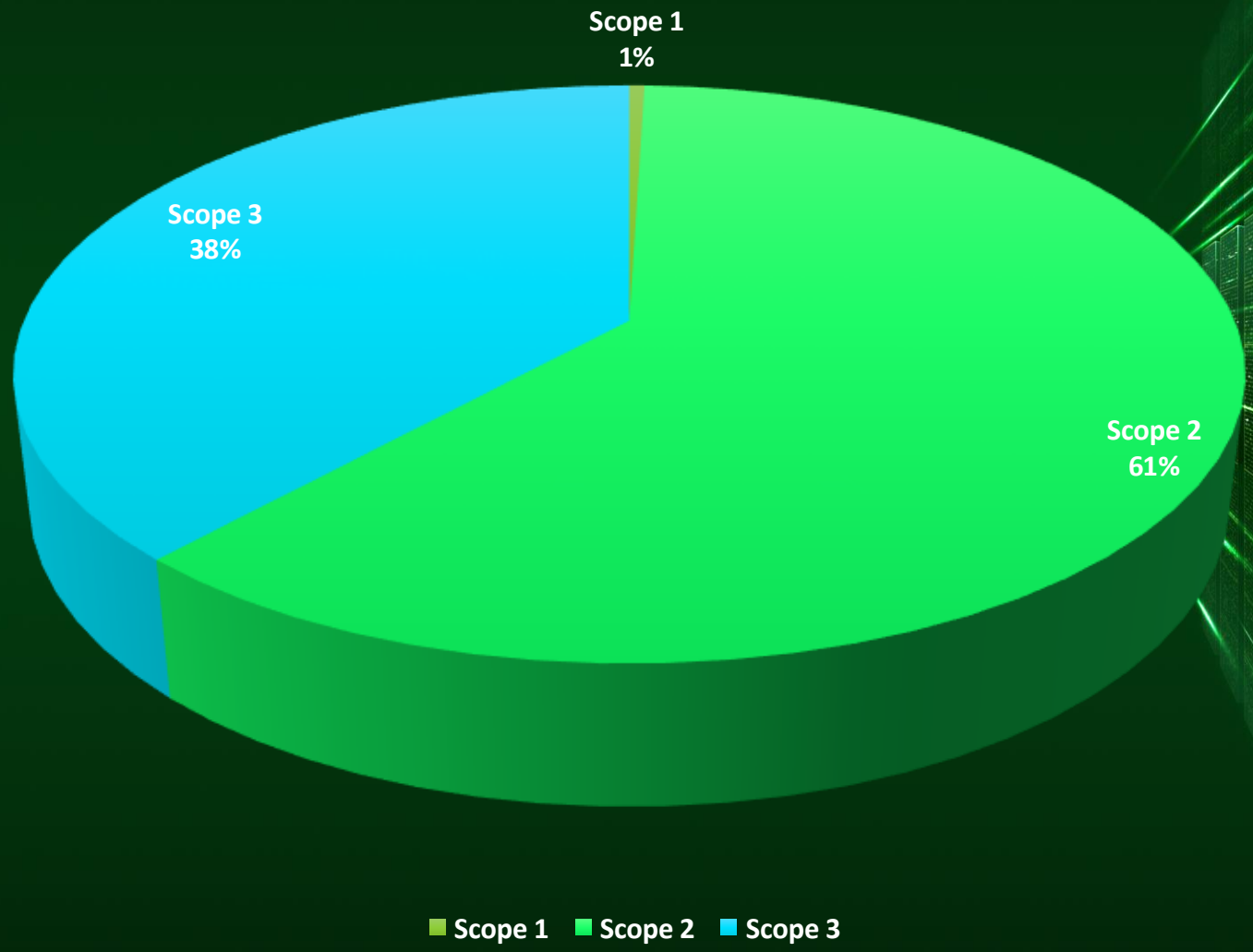
Data Center Carbon Footprint

- **Scope 1:**
- Diesel backup generators
- Refrigerant leaks
- On-site fossil fuel heating

- **Scope 2:**
- Purchased electricity for operation
- Purchased cooling and chilled water

- **Scope 3:**
- ICT/IT equipment (servers/GPUs)
- Electrical and MEP infrastructure
- Core and shell buildings

Cumulative total carbon footprint profile for 20 years



P. Lin, R. Bunger, V. Avelar, Quantifying data center Scope 3 GHG emissions to prioritize reduction efforts.

What Suppliers Currently Provide

- **Lifecycle Coverage:** EPDs report impacts across Cradle-to-Gate (A1-A3), Installation (A4-A5), Use Phase (B6), and End-of-Life (C1-C4).
- **Embodied Focus:** EPDs are highly effective for embodied carbon (Scope 3 Category 1), but struggle with operational modeling.
- **Standard Formats:** Based on ISO 14025, 14040, 14044 standards, or PCRs from EN 50693, EPD Italy, PEP Ecopassport and ITU-TL.1410.
- **The Gap:** Use phase (B6) operational data is based on static assumptions that rarely match real-world data center deployment conditions.

ABB

Impact category	Unit	Total	Cradle-to-gate		Cradle-to-grave			
			UPSTREAM	CORE	DOWNSTREAM			
			Manufacturing		Distribution	Installation	Use and maintenance	End-of-life
GWP – total	kg CO_2 eq.	3.86E+03	1.50E+03	3.74E+02	8.79E+00	1.69E+02	1.79E+03	2.05E+01
GWP – fossil	kg CO_2 eq.	3.82E+03	1.49E+03	5.12E+02	8.78E+00	3.52E+00	1.78E+03	1.70E+01
GWP – biogenic	kg CO_2 eq.	3.44E+01	-3.37E+00	-1.39E+02	3.12E-03	1.65E+02	7.49E+00	3.53E+00

Use

The use stage considers the power losses over the reference service life as defined in the functional unit. This is calculated using the following formula, according to PCR EPDItaly007:

$$E_{use}[kWh] = \frac{P_{use} * 8760 * RSL * \alpha}{1000} = \frac{21.03\ \text{W} * 8760\ \text{hours} * 20\ \text{years} * 100\%}{1000} = 3684.46\ kWh$$

Where:

- E_{use} = Total energy use over the reference service life
- P_{use} = Reference power consumption in watts per 10 feet of reference product with 30% load rate
- RSL = Reference Service Life in years
- α = Use time rate of 100%
- 8760 is the number of hours in a year
- 1000 is the conversion factor from W to kW

	Dataset	Amount	Unit	Represent.
Energy	*Electricity, medium voltage {US}| market group for electricity, medium voltage | Cut-off, S*	0.486	kg CO_2-eq./kWh	United States

NEPD-14587-15176 ABB Outdoor Voltage Transformer VRO-150G

Use Phase Calculation

$$E_{\text{use}} = P_{\text{use}} \times h_{\text{annual}} \times RSL \times EF_{grid}$$

- Where:
- E_{use} = Total Scope 2 operational emissions (kg CO_2-eq).
- P_{use} = Equipment operational power consumption or power dissipation (kW).
- h_{annual} = Annual operating hours (hours/year, typically 8,760 h/yr for 24/7 continuous operation).
- RSL = Reference Service Life / equipment lifespan (years).
- EF_{grid} = Grid emission factor (kg CO_2-eq/kWh).

Root Causes of Information Gap

Varied System Boundaries

$$E_{\text{use}} = P_{\text{use}} \times h_{\text{annual}} \times RSL \times \text{EF}_{\text{grid}}$$

Omitted Use Phase

Many equipment EPDs classify as “cradle-to-gate” and the use stage as "Not Declared" (MND), forcing operators to guess operational losses.

Static Load Factors

EPDs calculate use phase losses based on PCR-regulated load factor, whereas data centers operate equipment in redundant pairs at different load factors.

Generic Energy Mix

Suppliers use static regional grid averages (e.g., US average), ignoring individual operator specific conditions: carbon-free energy (CFE) matching and local power source.

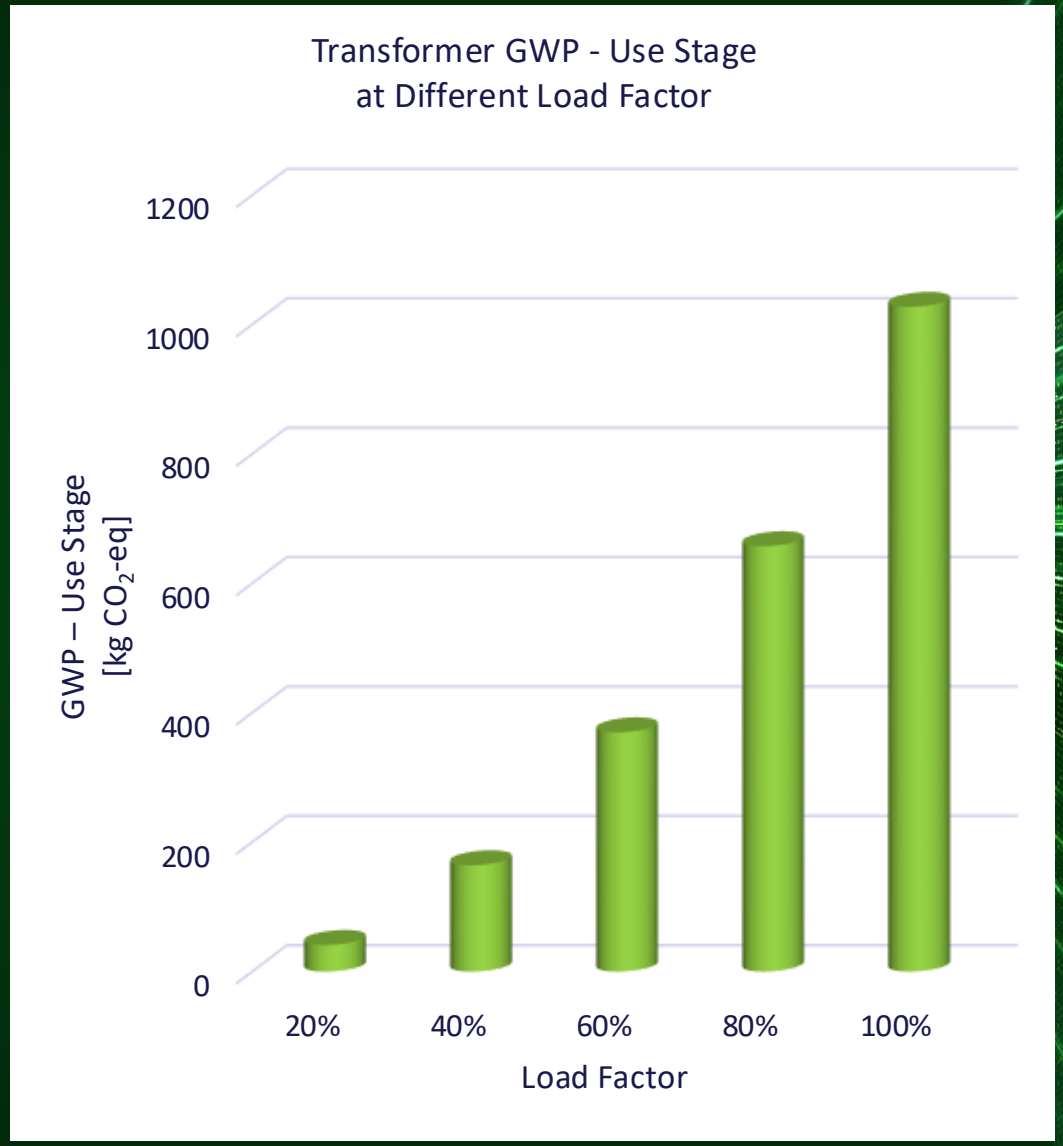
Various Life Span

Equipment life spans vary across categories: ICT ~2-5 years, capital goods ~20 years, building infrastructure up to 40~50 years.

Load Factor Realities

$$P_{\text{loss}}(t) = I(t)^2 R + P_{\text{no-load}}$$

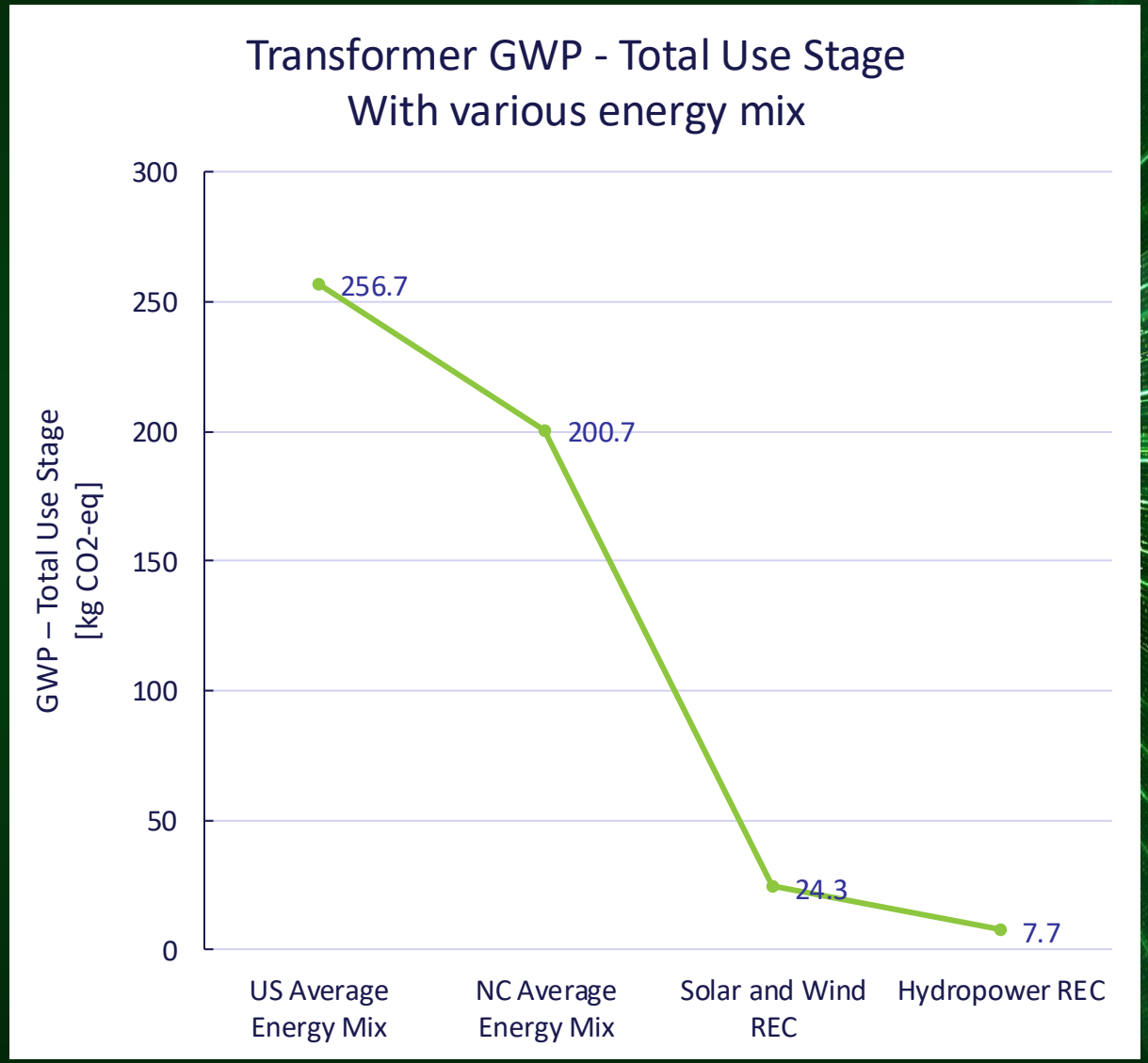
- **Loss Calculation:** Operational energy loss in conductors and distribution equipment is dominated by Joule heating, which scales quadratically with current.
- **Static EPD Approach:** PCRs generally assume a constant, flat operational load factor—typically 30% or 50% continuous nominal current over the 20–30 year Reference Service Life (RSL).
- **Dynamic Load Profile in DC:** Actual DC service experience highly variable loading (e.g., redundant N+1 data center architectures operating at 15–25% average load, vs. continuous heavy manufacturing at 80–90% load).



NEPD-14587-15176 ABB Outdoor Voltage Transformer VRO-150G

Discrepancy in Energy Mix

- Energy mix selection directly dictates the use-stage carbon footprint.
- Product EPDs rely on static national or regional grid averages, failing to account for site-specific grid decarbonization or time-of-use variations.
- Data centers require local energy mixes and RECs, making raw energy loss data (kWh) decoupled from pre-baked emission factors essential



Operational Calculation Mismatch

400%

Discrepancy Margin

Impact of Static Assumptions

Using static PCR values instead of dynamic, load-adjusted load factors can introduce significant error margin when calculating Scope 2 and Scope 3 DC operational carbon footprint.

This magnitude of discrepancy compromises regulatory compliance filings and ESG reporting.



Lifespan and DC Duty Cycle Divergence

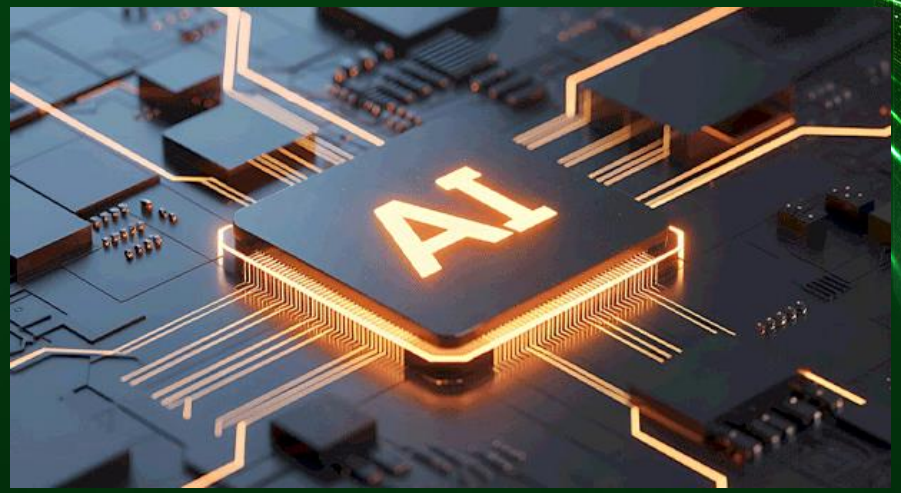
Electrical Infrastructure

- No-load core losses occur 24/7 regardless of server utilization.
- Load-dependent copper losses scale quadratically with current.
- Operators require parametric loss functions rather than flat annual kWh figures to model dynamic facility loads.



Servers, Storage & AI Accelerators

- Dynamic Power Ranges: AI server nodes swing from 200W idle to 1200W+ under peak matrix multiplication workloads.
- Short Lifespans: 3-year GPU hardware cycles increase embodied carbon per compute unit.
- EPDs often omit dynamic workload-dependent power curves.



EPD Limitations vs DC Operator Needs

Parameter	Standard Supplier EPD Baseline	Data Center Operator Requirement	Gap Impact
Operating Load	Static 50–100% continuous load	Dynamic 20–50% redundant load curve	Underestimates thermal losses
Energy Mix	Static national/regional grid average	Decoupled kWh loss + renewable energy mix	Distorts Scope 2 accounting
Lifespan	Fixed 15–20 year assumption	Parametric annualized or duty-cycle based	Misallocates annual Scope 3

EPD Harmonization

Adopt Open Standards

Align with established industrial standards and iMasons Climate Accord and Open Compute Project (OCP) carbon disclosure guidelines.

Disclose Multipoint Losses

Provide energy loss data across 10%, 25%, 50%, 75%, and 100% load points in EPD Module B6.

Decoupled Energy Data

Mandate clear separation between pure energy loss metrics (kWh) and energy mix carbon factors, allowing operators to apply custom CFE emission intensity.

Call to Action

- Drive industry awareness: establish the OCP sub-project “Intelligent Lifecycle Management” to investigate operational carbon accounting gaps.
- Unite key stakeholders: align DC operators, MEP/ICT infrastructure suppliers, and energy providers on common EPD requirements.
- Publish a joint white paper: propose an industry-wide EPD harmonization framework to standardize reporting methodologies.

OCP – Intelligent Lifecycle Management

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Thank You!



Lifespan and DC Duty Cycle Divergence

Lifespan Mismatches: EPDs assume standard 15–20 year constant lifespans for electrical gear, whereas ICT server hardware refreshes every 3–5 years, invalidating static lifetime energy totals.

Operating Mode Variations: Modern UPS systems dynamically switch between Double Conversion (94% efficiency) and Eco-Mode (99% efficiency). EPDs assume a single fixed mode.

Thermal Operating Conditions: Laboratory EPD tests at 20°C ignore elevated ambient operating temperatures in high-density data center hot aisles, which alter electrical resistance losses.

Isolated Component Scenarios: EPDs assess components independently with default factors, failing to capture the real operating conditions.